

Latent coverage of noisy calibration thresholds

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SUMMARY

Prediction intervals for a latent quantity, such as the mean of an unsampled area, are often calibrated on noisy proxies. Calibration on noisy scores is conservative when the latent distribution and the noise are both symmetric and unimodal, but a symmetric latent distribution is seldom plausible. For bi-log-concave latent distributions, which include the log-concave ones and some multimodal ones, we show that little is lost without symmetry and quantify the loss for classes of noise laws, exactly or within certified bounds: noisy coverage 0.9 guarantees latent coverage 0.8991 under Gaussian noise, 0.8981 under any symmetric unimodal noise, heavy-tailed or not, and 0.8935 under any mean-zero log-concave noise, against 0.8 for median-zero noise without a shape assumption, whereas under mean-zero unimodal noise nothing can be guaranteed. Hence the noisy conformal threshold at a slightly raised rank covers the latent target with high probability whatever the noise variances, which may differ between units and be unknown; without the raise its latent reliability can tend to zero as the calibration sample grows. Under Gaussian noise the worst-case widening of the calibrated radius is governed by the location of the latent mean: with the mean at a fixed distance from the boundary of the interval it is of the order of the noise variance, even for asymmetric distributions with mean zero, and it reaches the order of the noise standard deviation only when the mean is within a bounded number of noise standard deviations of the boundary.

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Some key words: Conformal prediction; Deconvolution; Log-concavity; Measurement error; Small area estimation.

1. INTRODUCTION

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Suppose that a residual W , the latent target minus a centre fitted on independent training data, is observed only through $V = W + e$, where $e \sim N(0, D)$ is independent of W and, until § 4.2, D is known. A survey statistician who calibrates a prediction interval for the true mean of an unsampled area on the direct estimates of sampled areas is in this position (Fay & Herriot, 1979), as is a conformal analyst whose calibration responses carry measurement error (Einbinder et al., 2024; Cohen et al., 2025). The threshold t at which the noisy scores $|V|$ have coverage p is observable. We ask which radius gives the latent residual coverage q , uniformly over a class of latent distributions.

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If W is symmetric and unimodal, the theorem of Anderson (1955) gives $\text{pr}(|W + e| \leq t) \leq \text{pr}(|W| \leq t)$, so the noisy threshold is conservative (Einbinder et al., 2024). Without symmetry the available transfers make no shape assumption: a median-zero noise turns a noisy level $1 - \alpha$ into a latent level $1 - 2\alpha$ (Guille-Escuret & Ndiaye, 2024), and LatentCP (Zheng et al., 2026), with a marginal guarantee and no shape assumption, uses a Markov-type compatibility threshold. Cohen et al. (2025) estimate the noise-free threshold by deconvolution, without a latent guarantee;

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conformal intervals for small areas (Bersson & Hoff, 2024) cover observable responses rather than latent area means, and conformal confidence intervals (Zhang & Tuoto, 2026) target the expectations of sampled units rather than a new, unsampled one; empirical Bayes intervals (Armstrong et al., 2022; Ignatiadis & Wager, 2022) concern sampled units, and parametric small-area prediction intervals (Hall & Maiti, 2006; Yoshimori & Lahiri, 2014) rely on a model for the latent effects. For a fixed smooth latent distribution, Gaussian noise moves quantiles by $O(D)$ (Jochmans & Weidner, 2024); a guarantee, however, must hold for the worst distribution in a class, and there the order and its cause are different.

Uniformly over a shape class the picture is organized by the location of the mean. We work with log-concave laws and with the larger class of bi-log-concave laws (Dümbgen et al., 2017), which does not require unimodality. Without a restriction on the mean the worst-case widening at equal levels is at most $c_q D^{1/2}$, with a constant that is sharp as $D \rightarrow 0$ (Theorem 1); if the mean lies at distance at least d from the boundary of the interval the widening is at most $D/(2d)$ for every D , an order attained by asymmetric laws with mean zero (Theorem 2 and Supplementary Proposition S1); and a widening of order $D^{1/2}$ requires the mean within $O(D^{1/2})$ of the boundary, with a limiting coefficient given by a one-dimensional extremum (Theorem 3). The widening is also small, at most 1.16% of the radius at $q = 0.9$, and the Gaussian law of the noise matters little: § 4.3 gives the transfer, exact at each end, for any class of noise laws closed under scaling and negation, explicitly for symmetric unimodal noise and, with a certificate, for mean-zero log-concave noise. Measured in coverage, noisy coverage 0.9 implies latent coverage 0.8991 at every noise level, where the shape-free transfer of Guille-Escuret & Ndiaye (2024) gives 0.8 for median-zero noise, and 0.8992 cannot be guaranteed (Corollary 1). So a slack of one part in a thousand absorbs the widening for every noise variance at once, and the noisy threshold at a suitable rank is valid for the latent target when the noise variances are heterogeneous and unknown (Proposition 3). Without the slack the guarantee is not merely slightly off: for some log-concave laws the latent reliability of the usual high-probability rank tends to zero as the calibration sample grows (Proposition 4), which matters for conformal calibration on large sets of noisy regression responses more than for small areas.

2. THE SHARP RADIUS

Let $\mathcal{L}$ be the class of log-concave laws on the real line, including point masses, and $\mathcal{B} \supset \mathcal{L}$ the class of laws that are point masses or have a continuous distribution function F with $\log F$ and $\log(1 - F)$ concave. Both classes are closed under translation and scaling, and convolution with a log-concave law preserves them (Prékopa, 1973); the Supplementary Material treats $\mathcal{B}$. After scaling t to 1, write $x = D/t^2$ and let $Z \sim N(0, 1)$ be independent of W . Define

$$R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{L}, \text{pr}(|W + x^{1/2}Z| \leq 1) \geq p\}, \quad (1)$$

where Q_q is the lower q -quantile, and $R_{p,q}(x) = -\infty$ if nothing is feasible, which happens exactly when $\text{pr}(|x^{1/2}Z| \leq 1) < p$. If calibration certifies noisy coverage at least p at threshold t , then $t R_{p,q}(D/t^2)$ is the smallest radius that guarantees latent coverage q for every log-concave latent law; this is sharpness within the class and for one constraint, not optimality among procedures that use the whole calibration sample. We call $R_{p,q}(x) - 1$ the radius correction. Let $R_{p,q}^{\mathcal{B}}$ be defined by (1) with $\mathcal{B}$ in place of $\mathcal{L}$, so $R_{p,q}^{\mathcal{B}} \geq R_{p,q}$. The upper bounds of §§ 3–4.1 hold for $R_{p,q}^{\mathcal{B}}$; the exact radius $R_{p,q}$ at a fixed x and its computation use log-concavity. For $0 \leq b \leq 1$ let $R_{p,q}^{(b)}(x)$ be defined by (1) with the additional constraint $|E(W)| \leq b$, so that $1 - b$ bounds the distance of the mean from the boundary.

Write $g_x(w) = \text{pr}(|w + x^{1/2}Z| \leq 1)$, so that the constraint in (1) reads $E\{g_x(W)\} \geq p$, and let $\mathcal{F}$ be the family of point masses and of laws with density proportional to $e^{\beta w}$ on a segment $[a, b]$.

PROPOSITION 1. $R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{F}, E\{g_x(W)\} \geq p\}$. The same holds for any continuous constraint kernel that vanishes at $\pm\infty$, in particular for the average kernel of § 4.2. 85

The proof uses the localization theorem of Fradelizi & Guédon (2004) (Supplementary Material); over $\mathcal{F}$ the problem is a two-dimensional maximization of closed forms.

3. WORST-CASE RADIUS CORRECTION AND THE LOCATION OF THE MEAN

3.1. A one-sided problem 90

Near each end of the interval the problem is one-sided; at the right end $Y = (W - 1)/x^{1/2}$ rescales it. For $u > 0$ let E_u be exponential with rate u , independent of Z , and for the exponential tail $b - E_u$ let

$$m_u(b) = \text{pr}(b - E_u + Z \leq 0) = \Phi(-b) + e^{-ub+u^2/2} \Phi(b - u),$$

which is continuous and strictly decreasing in b . Let $b_u(p)$ solve $m_u(b) = p$, and write $v_q(u) = b_u(q) - u^{-1} \log(1/q)$ and $k_q(u) = u^{-1} - b_u(q)$. The law $b_u(q) - E_u$ has q -quantile $v_q(u)$ and mean $-k_q(u)$. 95

LEMMA 1. Let $c_{p,q} = \sup\{F_Y^{-1}(q) : Y \in \mathcal{L}, \text{pr}(Y + Z \leq 0) \geq p\}$ and $c_q = c_{q,q}$. Then $c_{p,q} = \sup_{u>0}\{b_u(p) - u^{-1} \log(1/q)\}$, $c_q > 0$, and $q \mapsto c_q$ is nonincreasing. The same holds with $\mathcal{B}$ in place of $\mathcal{L}$.

LEMMA 2. Let $e^{-1} < q < 1$ and $\kappa \geq 0$. Then 100

$$\sup\{F_Y^{-1}(q) : Y \in \mathcal{B}, E(Y) \leq -\kappa, \text{pr}(Y + Z \leq 0) \geq q\} = L_q(\kappa) = \sup\{v_q(u) : u > 0, k_q(u) \geq \kappa\}.$$

The function L_q is positive and nonincreasing, and, for $q > 1/2$, $L_q(\kappa) = c_q$ for $\kappa \leq \kappa_q^*$, where κ_q^* is the largest value of k_q on the compact set of maximizers of v_q , and $\kappa L_q(\kappa) \rightarrow \{1 - \log(1/q)\}/2$ as $\kappa \rightarrow \infty$.

Exponential tails are extremal because concavity of $\log F_Y$ limits how fast mass can accumulate below the q -quantile; the proofs replace Y by the exponential tail through a supergradient of $\log F_Y$ at that quantile, which keeps the quantile, the constraint and the bound on the mean. Ball arithmetic gives $c_{0.8} = 0.04607$, $c_{0.9} = 0.01906$ and $c_{0.95} = 0.00841$, and in double precision $\kappa_q^* = 6.81, 20.18$ and 50.07 . 105

3.2. No restriction on the mean 110

THEOREM 1. For every $q \in (0, 1)$, every $x > 0$ and every $W \in \mathcal{B}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$, $Q_q(|W|) \leq 1 + c_q x^{1/2}$. Moreover $R_{q,q}(x) = 1 + c_q x^{1/2} + o(x^{1/2})$ as $x \rightarrow 0$, and likewise for $R_{q,q}^{\mathcal{B}}$. 110

Proof of the upper bound. Let $\alpha_R = \text{pr}(W + x^{1/2}Z > 1)$ and $\alpha_L = \text{pr}(W + x^{1/2}Z < -1)$, so that $\alpha_L + \alpha_R \leq 1 - q$. The law of $Y = (W - 1)/x^{1/2}$ has $\log F_Y$ concave and $\text{pr}(Y + Z \leq 0) = 1 - \alpha_R$. By Lemma 1, $F_Y^{-1}(1 - \alpha_R) \leq c_{1-\alpha_R} \leq c_q$, that is, $\text{pr}(W > 1 + c_q x^{1/2}) \leq \alpha_R$. The same argument applied to $-W$, which uses concavity of $\log(1 - F_W)$, gives $\text{pr}(W < -1 - c_q x^{1/2}) \leq \alpha_L$, so $\text{pr}(|W| \leq 1 + c_q x^{1/2}) \geq q$. 115 $\square$

The lower bound uses boundary-layer laws $W = 1 + x^{1/2}\{b_u(q + \eta_x) - E_u\}$, whose right end point lies just outside the threshold (Supplementary Material). Since $E\{g_x(W)\} \leq g_x(0)$, the constraint is feasible only if $x^{1/2} \leq 1/\Phi^{-1}\{(1+q)/2\}$, so Theorem 1 gives $Q_q(|W|) - 1 \leq$

¹²⁰ $c_q/\Phi^{-1}\{(1+q)/2\}$, which is below 1.16% at $q = 0.9$ for every $W \in \mathcal{B}$. For log-concave laws, double-precision upper bounds give $R_{q,q}(x) \leq 1$ for $x \geq 0.013$, hence a correction below 0.22% (Supplementary Material).

3.3. The mean away from the boundary

¹²⁵ **THEOREM 2.** *Let $W \in \mathcal{B}$ with $|E(W)| \leq B$, let $\sigma > 0$ and $t > B$, and suppose that $\text{pr}(|W + \sigma Z| \leq t) \geq q$. Then*

$$Q_q(|W|) \leq B + \{(t - B)^2 + \sigma^2\}^{1/2} \leq t + \frac{\sigma^2}{2(t - B)}.$$

In particular $R_{p,q}^{(b)}(x) \leq 1 + x/\{2(1 - b)\}$ for all $p \geq q$ and $x > 0$, and $Q_q(|W|) \leq (t^2 + \sigma^2)^{1/2}$ if $E(W) = 0$.

¹³⁰ A mean far from an end of the interval caps how steeply the density can rise towards it; the proof makes this precise with the heat equation for quantiles (Steinbrecher & Shaw, 2008), which gives $d\{r(s) - B\}^2/ds \geq -1$ for $r(s) = Q_q(|W + s^{1/2}Z|)$. It improves on Theorem 1 only when $x^{1/2} < 2c_q(1 - b)/(1 - c_q^2)$; for mean-zero laws at $q = 0.9$ this means $x < 1.5 \times 10^{-3}$, where both bounds are below 0.08%. The order x cannot be improved, even for mean-zero laws: centred log-affine laws on a segment attain it, with coefficient up to 0.0747 at $q = 0.9$, numerically (Supplementary Proposition S1), so centring does not replace symmetry.

3.4. The transition

¹³⁵ **THEOREM 3.** *Let $1 - e^{-1} < q < 1$, $x > 0$ and $W \in \mathcal{B}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$. Then $|E(W)| < 1$, and if $\Delta = Q_q(|W|) - 1 > 0$,*

$$2\{1 - |E(W)|\}\Delta + \Delta^2 \leq x. \quad (2)$$

Moreover, for every $\kappa \geq 0$,

$$\lim_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} = L_q(\kappa), \quad b_x = 1 - \kappa x^{1/2},$$

and the same limit holds with $\mathcal{L}$ replaced by $\mathcal{B}$ in the definition of $R^{(b)}$.

¹⁴⁰ In the original units the mean lies $\kappa D^{1/2}$ from the threshold, and the unrestricted worst case $c_q D^{1/2}$ persists until κ exceeds κ_q^* , about 20 noise standard deviations at $q = 0.9$; beyond it the coefficient decays like $\{1 - \log(1/q)\}/(2\kappa)$. Proofs and a figure are in the Supplementary Material.

4. CALIBRATION SLACK AND A FINITE-SAMPLE PROCEDURE

4.1. Slack removes the correction

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The proof of Theorem 1 splits the noisy failure budget between the two ends. With a noisy level $p \geq q$, splitting both the noisy and the latent budgets gives, for every x and every $W \in \mathcal{B}$ feasible at level p , $Q_q(|W|) \leq 1 + C_{p,q}x^{1/2}$, where $C_{p,q} = \sup_{\alpha_L + \alpha_R = 1-p} \inf_{\beta_L + \beta_R = 1-q} \max(c_{1-\alpha_L, 1-\beta_L}, c_{1-\alpha_R, 1-\beta_R})$, nonincreasing in p , is at least $c_{p,q}$.

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COROLLARY 1. *If $C_{p,q} \leq 0$ then $R_{p,q}^{\mathcal{B}}(x) \leq 1$ for every x , and if $c_{p,q} > 0$ then $R_{p,q}(x) > 1$ for all small x . At $q = 0.9$ the smallest slack $p - q$ with $C_{p,q} \leq 0$, and with $R_{p,q}(x) \leq 1$ for all x , lies in $(7.9 \times 10^{-4}, 10^{-3}]$; other levels are in the Supplementary Material. In coverage terms, noisy coverage 0.9 gives latent coverage 0.8991 for every $W \in \mathcal{B}$ and noise variance, while 0.8992 fails for some log-concave laws.*

4.2. Heterogeneous and unknown noise variances

Let $W_1, \dots, W_K, W_{\text{new}}$ be independent with law G , independent of $e_i \sim N(0, D_i)$ with the D_i fixed. The scores $V_i = W_i + e_i$ are independent but not identically distributed, and $\mathcal{D}$ denotes the calibration data.

PROPOSITION 2. *Let $T = |V|_{(k)}$, let $\bar{F}$ be the average distribution function of the $|V_i|$, assumed continuous, and let p_k be the δ -quantile of the Beta($k, K + 1 - k$) distribution. If $k \geq Kp_k + 1$ then $\text{pr}\{\bar{F}(T) \geq p_k\} \geq 1 - \delta$.*

The proof uses Hoeffding (1956).

PROPOSITION 3. *Let $G \in \mathcal{B}$, let the D_i be unknown, and suppose that $k \geq Kp_k + 1$ and $C_{p_k, q} \leq 0$. Then $\text{pr}_{\mathcal{D}}\{\text{pr}(|W_{\text{new}}| \leq T \mid \mathcal{D}) \geq q\} \geq 1 - \delta$.*

Proof. On the event of Proposition 2, $K^{-1} \sum_i \text{pr}(|W + e_i| \leq T) \geq p_k$ for $W \sim G$. If $\text{pr}(|W| \leq T) < q$, Corollary 1 at threshold T gives $\text{pr}(|W + e_i| \leq T) < p_k$ for every i , whatever D_i , and so for the average, a contradiction. $\square$

At $q = 0.9$ and $\delta = 0.05$ the conditions hold at the smallest rank with $p_k \geq 0.901$ and $k \geq Kp_k + 1$, which exists for $K \geq 29$ and, for $K \leq 3000$, is at most three ranks above the usual high-probability rank, the smallest with $p_k \geq 0.9$ (Supplementary Material). For $K = 110$ the two coincide, at $k = 105$ with $p_k = 0.9068$. The marginal rank $\lceil 0.9(K + 1) \rceil = 100$ is not covered, and for large K neither the marginal nor the usual rank is valid for the latent target in general (Proposition 4).

PROPOSITION 4. *Let $q \in (0, 1)$. There are a log-concave law G and $D > 0$ such that, with $D_i = D$ for all i , $\text{pr}_{\mathcal{D}}\{\text{pr}(|W_{\text{new}}| \leq |V|_{(k_K)} \mid \mathcal{D}) \geq q\} \rightarrow 0$ as $K \rightarrow \infty$ for every sequence of ranks with $k_K/K \rightarrow q$, such as the marginal rank and, for any fixed δ , the usual high-probability rank.*

Proof. Since $c_q > 0$, Theorem 1 gives $R_{q, q}(x) > 1$ for some x , so some log-concave W has $H(1) \geq q$ and $r_q = Q_q(|W|) > 1$, where H is the distribution function of $|W + x^{1/2}Z|$; take G the law of W and $D = x$. Latent coverage at $T = |V|_{(k)}$ is at least q if and only if $T \geq r_q$, that is, if and only if fewer than k of the $|V_i|$ are below r_q , a $\text{Bin}\{K, H(r_q)\}$ count. As H is strictly increasing on $(0, \infty)$, $H(r_q) > H(1) \geq q$, so $\text{pr}\{\text{Bin}(K, H(r_q)) \leq k_K - 1\} \rightarrow 0$ by the law of large numbers. $\square$

The proof also gives the exact latent reliability of rank k with $D_i = D$, $\text{pr}\{\text{Bin}(K, H(r_q)) \leq k - 1\}$. At the stress-test law of the Supplementary Material, with $D^{1/2}$ equal to 1% of the noisy q -quantile, the usual rank has latent reliability 0.943, 0.918 and 0.151 at $K = 10^3, 10^4$ and 10^6 , against 0.954, 0.958 and 0.989 for the certified rank. For $29 \leq K \leq 300$ the shortfall is at most 0.005; for larger K the reliability collapses, while the latent coverage tends to 0.89918.

If the variances have a known lower bound $D_{\min}$, the same guarantee holds for the shrunk threshold $\max\{0, T + C_{p', q} D_{\min}^{1/2}\}$ whenever $q \leq p' \leq p_k$ and $C_{p', q} < 0$; at $q = 0.9, K = 110$ and $\delta = 0.05$ this is $T - 0.114 D_{\min}^{1/2}$, while using the smallest variance as a common kernel variance is not valid (Supplementary Material). The sharpest use of known variances is the average kernel $\bar{g}(w) = K^{-1} \sum_i \text{pr}(|w + e_i| \leq T)$: $\int \bar{g} dG \geq p_k$ on the event of Proposition 2, so by Proposition 1 the computed radius $s = TR^{\text{mix}}$, with R^{mix} defined by (1) with the constraint $\int \bar{g} dG \geq p_k - \omega$ and ω bounding a discretization error, has the same guarantee for log-concave G .

For the computed radius our implementation returns a numerical upper bound on R^{mix} , by a branch and bound in double precision, not a certificate.

4.3. *Non-Gaussian noise*

The Gaussian law of the noise enters the transfer only through its constants. Let $\mathcal{E}$ be a class of noise laws closed under scaling and negation, and let

$$\Psi_{\mathcal{E}}(q) = \sup_{\epsilon \in \mathcal{E}} E\{\min(1, qe^{-\epsilon})\}.$$

200 **PROPOSITION 5.** *Let ϵ have a law in $\mathcal{E}$ and be independent of Y , with $\log F_Y$ concave. If $\text{pr}(Y + \epsilon \leq 0) \geq p > \Psi_{\mathcal{E}}(q)$, then $F_Y(0) \geq q$. If $p < \Psi_{\mathcal{E}}(q)$, some exponential tail Y with $F_Y(0) < q$ and some ϵ with law in $\mathcal{E}$ have $\text{pr}(Y + \epsilon \leq 0) > p$.*

Proof. If $v = F_Y^{-1}(q) > 0$, the exponential tail $v + a/u - E_u$, $a = \log(1/q)$, u a supergradient of $\log F_Y$ at v , lies stochastically below Y , and adding ϵ preserves this, so $p \leq \text{pr}(a/u - E_u + \epsilon \leq 0) = E\{\min(1, qe^{-u\epsilon})\} \leq \Psi_{\mathcal{E}}(q)$, as $u\epsilon$ has a law in $\mathcal{E}$. If F_Y has an atom at v with $Y \geq v$, as
205 for a point mass, then $p \leq \text{pr}(\epsilon < 0) \leq \lim_{u \rightarrow \infty} E\{\min(1, qe^{-u\epsilon})\} \leq \Psi_{\mathcal{E}}(q)$. Conversely, $a - E_1$ with noise ϵ attains $E\{\min(1, qe^{-\epsilon})\}$; shift it slightly to the right. $\square$

Applied at both ends as in Corollary 1, it extends Proposition 3 to any noise laws in $\mathcal{E}$, possibly different between areas and with unknown scales: a rank whose p_k exceeds $p_{\mathcal{E}}$, or equals a
210 certified level, suffices, where $p_{\mathcal{E}}$ is the infimum of the noisy levels whose budget, however split between the ends, leaves latent budgets summing to at most $1 - q$ (an end with noisy budget α needs latent budget $1 - \lambda$, where $\Psi_{\mathcal{E}}(\lambda) = 1 - \alpha$); negation enters only through the left end.

THEOREM 4. *Let $q > e^{-1}$. (i) For symmetric unimodal noise, $\Psi_{\mathcal{E}}(q) = (1 + qe^{-r})/2$, where $r > 0$ solves $(1 + r)e^{-r} = (1 + \log q)/q$, attained by uniform noise. (ii) For mean-zero log-concave
215 noise, $\Psi_{\mathcal{E}}(q)$ is the supremum over centred log-affine laws on segments, and at $q = 0.9$ it lies in $[0.90517, 0.9052]$, the lower end given by the centred exponential law. (iii) For mean-zero unimodal noise, $\Psi_{\mathcal{E}}(q) = 1$.*

Part (i) holds because symmetric unimodal laws are scale mixtures of uniform laws, part (ii) by localization with the constraint $E(\epsilon) \geq 0$ and a ball-arithmetic branch and bound, and part
220 (iii) by a law with most of its mass spread far below $\log q$, balanced by a rare long excursion (Supplementary Material).

At $q = 0.9$ the certified levels are $p_{\mathcal{E}} \in (0.90079, 0.901]$ for Gaussian noise, $(0.9017, 0.9018]$ for symmetric unimodal noise, heavy-tailed or not, and $(0.90517, 0.906]$ for mean-zero log-concave noise, skewed or not; noisy coverage 0.9 guarantees latent coverage 0.8991, 0.8981
225 and 0.8935. So the noisy threshold at rank 105 of Table 1, with $p_k = 0.9068$, is valid for every mean-zero log-concave measurement error, whatever its shape and variance in each area; by (iii), some such restriction is needed.

5. NUMERICAL ILLUSTRATION

The target is $\text{pr}_{\mathcal{D}}\{\text{pr}(W_{\text{new}} \in C \mid \mathcal{D}) \geq 0.9\} \geq 0.95$, with $K = 110$, $\text{var}(W) = 1$ and heterogeneous noise variances $D_i = 0.577 L_i / E(L_i)$, L_i lognormal with log-scale 0.7. Latent residuals,
230 all with mean zero, are drawn directly; conditional coverage given each data set is exact, and reliability is estimated from 150 data sets per law. Besides four standard log-concave laws we use a truncated exponential law in $\mathcal{F}$, a bimodal law in $\mathcal{B}$ but not in $\mathcal{L}$, and a t_3 law outside $\mathcal{B}$. We compare the three rules of § 4.2, a Fay–Herriot tolerance interval adapted from Chatterjee et al. (2008), and the Markov rule, the radius r with $\bar{g}(r) = \{p_k - q\bar{g}(0)\}/(1 - q)$, the smallest the single constraint allows without a shape assumption, at its best rank, 107 (Supplementary Material).

Table 1. *Reliability, times 100, and mean width (in parentheses); target 95*

latent law	Markov	Fay–Herriot	noisy threshold	shrunk	computed
normal	100 (5.92)	96 (3.93)	100 (4.96)	100 (4.90)	98 (4.45)
Laplace	100 (6.29)	93 (3.96)	100 (5.15)	100 (5.09)	99 (4.66)
centred gamma	100 (6.16)	100 (3.95)	100 (4.97)	100 (4.90)	100 (4.45)
truncated Laplace	100 (5.89)	85 (3.89)	100 (4.93)	100 (4.87)	97 (4.42)
truncated exponential	100 (6.17)	95 (3.93)	100 (5.06)	100 (4.99)	100 (4.55)
bimodal ($\mathcal{B}$ only)	100 (5.77)	100 (4.01)	100 (4.87)	100 (4.81)	99 (4.35)
t_3 (outside $\mathcal{B}$)	100 (5.70)	97 (3.80)	100 (4.66)	100 (4.60)	98 (4.09)

Noisy threshold, T at rank 105; shrunk, $\max\{0, T - 0.114D_{\min}^{1/2}\}$ with $D_{\min} = \min_i D_i$; computed, TR^{mix} ; centred gamma with shape 2; truncated exponential, density proportional to e^{4w} on $[0, 1]$, standardized; bimodal, the mixture of Supplementary Lemma S1; widths are full widths. The Markov rule is guaranteed for every law; under the bimodal law the noisy and shrunk thresholds are guaranteed and the computed radius is exploratory; under t_3 the noisy threshold is guaranteed by Anderson’s theorem. Standard errors are at most 2.9 for reliabilities and 0.08 for widths.

The rules with a guarantee reached the target for every law in their class, and the computed radius did so also under the bimodal and t_3 laws, for which it has none. The shrunk threshold was 1.2–1.4% shorter than the noisy threshold, and the computed radius 9.5–12% shorter than the noisy threshold and 24–28% shorter than the Markov rule. The Fay–Herriot interval was 7–15% shorter than the computed radius but missed the target under the truncated Laplace law by 5.8 standard errors; its other shortfalls are within Monte Carlo error. With 2000 data sets per law the noisy threshold, the shrunk threshold and the Markov rule reached conditional coverage 0.9 in every data set, with mean conditional coverage 0.97–0.99 for the first two: the guarantees are conservative for laws far from the extremal ones. In the Supplementary Material LatentCP (Zheng et al., 2026) is 16–18% longer than the computed radius, the deconvolution threshold of Cohen et al. (2025) has no latent guarantee, and at oracle-matched reliability the gains lie in the cost of a guarantee, not in the shape of the interval.

6. DISCUSSION

The guarantees are conditional on the calibration sample and assume noise independent of the latent residual, Gaussian or, for the noisy threshold, in one of the classes of § 4.3, and a bi-log-concave, or for the computed radius log-concave, latent law shared by the calibration areas and the new one. The target is an unsampled area; for a sampled area the direct estimate is itself informative, and intervals for the expectations of sampled units are a different problem (Zhang & Tuoto, 2026). The shared law is the substantive assumption. In the Supplementary Material a shift of the new residual by half a latent standard deviation lowered mean coverage by up to 0.027 for laws like those of Table 1, but near the extremal law a tenth of a standard deviation breaks the guarantee, and residual dispersion varying with the noise variance broke reliability given a new area’s variance; no guarantee covers these cases. Bi-log-concavity cannot be checked on the latent scale, but convolution with log-concave noise preserves it, so with equal variances a confidence band for the signed noisy residuals containing no bi-log-concave distribution function (Dümbgen et al., 2017) rules it out, though not conversely, and it excludes heavy tails, although the symmetric t_3 law of Table 1 is covered by Anderson’s theorem. The mean condition of Theorems 2 and 3 concerns the latent residual law, which an intercept in the fitted regression does not centre. The mean is identified, as $E(V) = E(W)$, but Theorem 2 improves the radius only where both bounds are below 0.08% of it.

Open questions include an analytic proof that the centred exponential law is extremal among mean-zero log-concave noise laws, the leading coefficient for centred laws, and whether $C_{p,q} = c_{p,q}$ and $R_{p,q}^{\mathcal{B}} = R_{p,q}$.

DECLARATION OF THE USE OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

[To be completed by the author.]

SUPPLEMENTARY MATERIAL

The Supplementary Material contains the omitted proofs, the certificates, further simulations and an application. Code reproducing every number is at <https://github.com/kunhailP/latent-coverage-noisy-thresholds>, tag biometrika-submission.

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